

Munkres 5

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August 2025

1 Introduction

Let $\{X_\alpha\}_{\alpha \in J}$ be an infinite collection of compact spaces, $X = \prod X_\alpha$. There exists a (special case) X and closed $A \subsetneq X$ such that $\pi_\alpha(A) = X_\alpha$ for all $\alpha \in J$.

Take $X = [0, 1]^\omega$, $F : X \rightarrow \mathbb{R}_{\geq 0}$ by $F(x) = \frac{1}{2}x_1 + \frac{1}{4}x_2 + \frac{1}{8}x_3 + \dots$. F is continuous since each $f_n : X \rightarrow \mathbb{R}_{\geq 0}$ given by $f_n(x) = \frac{1}{2^n}\pi_n(x)$ is continuous, and the sum of the f_n is continuous by Weierstrass M-test. Consider the closed set $A = \{x \in X : F(x) \leq \frac{1}{2}\}$. Note for $r \in I$, $x = (0, 0, \dots, r, 0, 0, \dots) \in A$ for all positions of r . Hence, $\pi_n(A) = I$ for all $n \in N$. Further, $F(\vec{1}) = 1$, so $\vec{1} \notin A$.

Tychonoff's theorem. Munkres's proof: Let $\{X_\alpha\}_{\alpha \in J}$ be a collection of compact spaces. Let $X = \prod X_\alpha$ and $\mathcal{A} \subset P(X)$ be a collection of closed sets with the finite intersection property. Let $\mathcal{D} \supset \mathcal{A}$ be maximal wrt FIP (elements of $\mathcal{D}$ need not be closed).

Fix $\alpha \in J$. Note $\overline{\pi_\alpha(\mathcal{D})} = \{\overline{\pi_\alpha(D)} \subset X_\alpha : D \in \mathcal{D}\}$ has the FIP. Pick $x_\alpha \in \bigcap \overline{\pi_\alpha(D)}$ by compactness of X_α . Let $x = (x_\alpha)_{\alpha \in J}$. Let $D \in \mathcal{D}$. Note $x_\alpha \in \overline{\pi_\alpha(D)}$, so for any $U_\alpha(x_\alpha)$, we have $U_\alpha \cap \pi_\alpha(D) \neq \emptyset$. Then there exists $y \in D$ for which $\pi_\alpha(y) \in U_\alpha$, or $y \in \pi_\alpha^{-1}(U_\alpha)$, so $D \cap \pi_\alpha^{-1}(U_\alpha) \neq \emptyset$. This holds for all D , so $\pi_\alpha^{-1}(U_\alpha) \in \mathcal{D}$. $\mathcal{D}$ is closed under finite intersections, so any basis element of x is contained in $\mathcal{D}$. It follows $x \in \overline{D}$ for all $D \in \mathcal{D}$.

Then

$$x \in \bigcap_{D \in \mathcal{D}} \overline{D} \subset \bigcap_{A \in \mathcal{A}} \overline{A} = \bigcap \mathcal{A}.$$

Let X be a space and $\mathcal{D}$ be maximal wrt FIP. (1) $x \in \overline{D}$ for all $D \in \mathcal{D}$ iff $\mathcal{U}_x \subset \mathcal{D}$. Let $U(x) \in \mathcal{U}_x$. For any D , $U \cap D \neq \emptyset$, so $U \in \mathcal{D}$ (by maximality of $\mathcal{D}$). Conversely, if $\mathcal{U}_x \subset \mathcal{D}$, fix $D \in \mathcal{D}$. Then for all $U(x) \in \mathcal{U}_x$, $U \cap D \neq \emptyset$ (by FIP), so $x \in \overline{D}$. (2) ?

A space X is said to have the countable intersection property if for all $\mathcal{A} \subset P(X)$ such that for any countable subcollection $\mathcal{A}' \subset \mathcal{A}$ such that $\bigcap \mathcal{A}' \neq \emptyset$, then $\bigcap \overline{\mathcal{A}} \neq \emptyset$. Show that X has the CIP is equivalent to X being Lindelof.

Suppose X has the CIP. Let $\mathcal{A}$ be an open covering of X . Then $\bigcap(X - \mathcal{A}) = \emptyset$, so for some countable $\mathcal{A}' \subset \mathcal{A}$ we have $\bigcap(X - \mathcal{A}') = \emptyset$, or $\bigcup \mathcal{A}' = X$. Conversely, suppose X is Lindelof. Let $\mathcal{A}$ have the CIP st $\bigcap \overline{\mathcal{A}} = \emptyset$, so $\bigcup(X - \overline{\mathcal{A}}) = X$. Pick countable $\mathcal{A}' \subset \mathcal{A}$ st $\bigcup(X - \overline{\mathcal{A}'}) = X$, so $\emptyset = \bigcap \overline{\mathcal{A}'} \supset \bigcap \mathcal{A}'$, cont.

Let X be compact Hausdorff and $x, y \in X$. Then x, y are connected iff x, y are quasicontinuous.

If x, y are connected, then x, y are quasicontinuous. Consider the reverse direction and define $\mathcal{A}$ the collection of all closed subspaces $A \subset X$ such that x, y are QC wrt A . Let $\mathcal{C} \subset \mathcal{A}$ be a chain. I (more so Munkres) claim $A = \bigcap \mathcal{C} \in \mathcal{C}$. Sps FTSOC $A = D \cup E$ where D, E are disjoint nonempty closed containing x, y , resp. Since X is normal, let $U(D), V(E)$ be disjoint open. Consider the chain $\mathcal{C} - U \cup V$. If every element is nonempty, by compactness of X , $\bigcap(\mathcal{C} - U \cup V) = \bigcap \mathcal{C} - U \cup V \neq \emptyset$, cont, since $A \subset U \cup V$. Hence for some $C \in \mathcal{C}$, $C \subset U \cup V$. Then $C \cap U, C \cap V$ is a separation of C containing x, y , resp, so x, y are not quasicontinuous wrt C , cont.

By Zorn's Lemma, since every chain $\mathcal{C} \subset \mathcal{A}$ has a $\subset$ -minimal element, $\mathcal{A}$ has a $\subset$ -minimal element, say A . Sps FTSOC A is not connected, so there are closed nonempty disjoint D, E fw $A = D \cup E$. Note x, y are in different components, otherwise x, y are not QC wrt A . WLOG $x, y \in D$. Then x, y are QC wrt $D \subsetneq A$, cont min of A .

Let $h : X \rightarrow Z$ be an embedding of X in compact Hausdorff Z . Then there exists a unique (up to equivalence) compactification Y of X with the property that there exists an embedding $H : Y \rightarrow Z$ st $H|_X = h$.

Note $h(X) \cong X$. Since Z is compact Hausdorff, $\overline{h(X)}$ is compact Hausdorff. Let space Y identify the points of $h(X)$ with those of X , so there exists a compact Hausdorff space $Y \supset X$ and a homeomorphism $H : Y \rightarrow \overline{h(X)}$ and $H|_X = h$. As a result $\overline{X} = H^{-1}(\overline{H(X)}) = Y$, so Y is a compactification of X . H extends to an embedding $H : Y \rightarrow Z$.

Conversely, suppose Y is a compactification of X with a corresponding H . Then H restricts to a homeomorphism $Y \rightarrow H(Y)$ and

$$H(Y) = H(\overline{X}) = \overline{H(X)}_{H(Y)} = \overline{h(X)}_{H(Y)}.$$

Since Y is compact, $H(Y)$ is compact and therefore closed in Z , so closure wrt $H(Y)$ is the same as closure wrt Z , and $H(Y) = \overline{h(X)}$. If $H_1 : Y_1 \rightarrow \overline{h(X)}, H_2 : Y_2 \rightarrow \overline{h(X)}$ are homeomorphisms fw $H_1|_X, H_2|_X = h$, then $K = H_2^{-1} \circ H_1 : Y_1 \rightarrow Y_2$ is a homeomorphism st for $x \in X$, $K(x) = H_2^{-1}(H_1(x)) = h^{-1}(h(x)) = x$, so Y_1, Y_2 are equivalent.

Let X be completely regular Hausdorff. X has a compactification Y for which any continuous bounded $f : X \rightarrow \mathbb{R}$ can be extended uniquely to $Y \rightarrow \mathbb{R}$.

Let J be an indexing set for the set of continuous bounded functions $X \rightarrow \mathbb{R}$. Suppose for $\alpha \in J$, $f_\alpha : X \rightarrow \mathbb{R}$ is bounded by $[a_\alpha, b_\alpha]$. Let $Z = \prod_{\alpha \in J} [a_\alpha, b_\alpha]$. By Tychonoff's theorem, Z is compact. Define $F : X \rightarrow Z$ by $F(x) = (f_\alpha(x))_{\alpha \in J}$. If $x \neq y$, since X is completely regular, there exists a bounded continuous maps $f_1, f_2 : X \rightarrow \mathbb{R}$ st $f_1(x) = 0, f_1(y) = 1, f_2(x) = 1, f_2(y) = 0$, so $F(x) \neq F(y)$, i.e. F is injective. Further, $U(x) \subset X$ is open, since X is completely regular, there exists bounded continuous $f_\alpha : X \rightarrow \mathbb{R}$ for which $f(X - U) = \{0\}$ and $f(x) = 1$. Then $N_x = F^{-1}(\pi_\alpha^{-1}(\mathbb{R}^+)) = (\pi_\alpha \circ F)^{-1}(\mathbb{R}^+) = f_\alpha^{-1}(\mathbb{R}^+) \subset U$, so F is an embedding.

By the earlier lemma, there exists a compactification Y of X st $H : Y \rightarrow Z$ is an embedding and $H|X = F$. Suppose $f : X \rightarrow \mathbb{R}$ is continuous bounded. Then there exists $\alpha \in J$ for which $f = f_\alpha$. Consider the continuous function $\pi_\alpha \circ H : Y \rightarrow Z$. Over X , $\pi_\alpha \circ H = \pi_\alpha \circ F = f_\alpha$, so $\pi_\alpha \circ H$ is an extension of f .

Now in general, suppose $f_1, f_2 : Y \rightarrow \mathbb{R}$ are extensions of some function st $f_1(y) \neq f_2(y)$ for some $y \in Y$. Pick disjoint $V_1(f_1(y)), V_2(f_2(y))$ and let $N_y = f_1^{-1}(V_1(f_1(y))) \cap f_2^{-1}(V_2(f_2(y)))$ be open. Since $y \in \bar{X}$, $N_y \cap X \neq \emptyset$, so let $x \in N_y \cap X$. Then $f_1(y) \in f_1(N_y) \subset f_1(f_1^{-1}(V_1)) \subset V_1$. Similarly, $f_2(y) \in V_2$, cont as $f_1(y) = f_2(y)$. If $f : X \rightarrow \mathbb{R}$ can be extended to a compactification $Y \rightarrow \mathbb{R}$ (which we showed earlier for this special case of Y), then the extension is unique. (Note from Munkres: compactness was not used here).

Let X be completely regular and Y have the property from above. Let C be compact Hausdorff and $f : X \rightarrow C$ be continuous. Then f extends uniquely to $g : Y \rightarrow C$.

Since C is compact Hausdorff, let $k : C \rightarrow [0, 1]^J$ be an embedding and $D = k(C) \cong C$. Consider the maps

$$X \xrightarrow{f} C \xrightarrow{k} [0, 1]^J \xrightarrow{\pi_\alpha} [0, 1]$$

given by $f_\alpha = \pi_\alpha \circ k \circ f : X \rightarrow [0, 1]$. Since Y has the extension property, there exists a continuous extension $f'_\alpha : Y \rightarrow \mathbb{R}$ of f . Since $f_\alpha(X) \subset [0, 1]$, we can take continuous $f''_\alpha = \max(\min(f'_\alpha, 1), 0) : Y \rightarrow [0, 1]$ to be an extension of f_α . Now define $F : Y \rightarrow [0, 1]^J$ by $F(x) = (f''_\alpha(x))_{\alpha \in J}$. If $x \in X$,

$$F(x) = ((\pi_\alpha \circ k \circ f)(x))_{\alpha \in J} = (k \circ f)(x) \quad (1).$$

Further, since F is continuous,

$$F(Y) = F(\bar{X}) \subset \overline{F(X)} \subset \bar{D} = D.$$

The last equality is from D being a compact subset of the compact Hausdorff space $[0, 1]^J$. Restrict $F : Y \rightarrow D$. Hence, we may restrict k to a homeomorphism $k : C \rightarrow D$ and $g = k^{-1} \circ F : Y \rightarrow C$ is continuous. Indeed, for $x \in X$,

applying (1), $g(x) = k^{-1}(k(f(x))) = f(x)$, so g is an extension of f . The uniqueness of g is for the same reasons as earlier.

Uniqueness of Y . Let X be completely regular and Y_1, Y_2 be compactifications as above. Then Y_1, Y_2 are equivalent. Consider the inclusion map $i_1 : X \rightarrow Y_2$. Since i_1 is continuous, Y_1 has the extension property, and Y_2 is compact Hausdorff, i_1 extends to some $f : Y_1 \rightarrow Y_2$. Similarly, there exists $g : Y_2 \rightarrow Y_1$ extending $i_2 : X \rightarrow Y_1$. Since $g \circ f : Y_1 \rightarrow Y_1$ and $\text{id}_{Y_1} : Y_1 \rightarrow Y_1$ are both extensions of i_2 , $g \circ f = \text{id}_{Y_1}$. Similarly, $f \circ g = \text{id}_{Y_2}$, so f, g are bijective and continuous inverses of each other and $f : Y_1 \rightarrow Y_2$ is our desired equivalence map.

Given completely regular X , the space $Y \supset X$ is denoted by $\beta(X)$, the Stone-Cech compactification of X .

The inclusion map $i : X \rightarrow Y$ is continuous. Since $\beta(X)$ has the extension property and Y is compact Hausdorff, i extends to continuous $g : \beta(X) \rightarrow Y$. Since $\beta(X)$ is compact, $g(\beta(X))$ is compact, and therefore closed (being a compact subset of compact Hausdorff Y). Since X is contained in $g(\beta(X))$, $Y = \overline{X} \subset g(\beta(X)) \subset Y$, so $g(\beta(X)) = Y$, i.e. g is surjective. If $A \subset \beta(X)$ is closed, A is compact since $\beta(X)$ is compact. Then $g(A)$ is compact and therefore closed. Since g is a closed surjective continuous map, g is a quotient map.

$\beta(X)$ is connected iff X is connected. If X is connected, $\beta(X) = \overline{X}$ is connected. Now suppose $X = U \cup V$ is a separation, U, V nonempty open (wrt X). Let $\{0, 1\}$ have the discrete topology. $f : X \rightarrow \{0, 1\}$ given by $f(U) = \{0\}$ and $f(V) = \{1\}$ is continuous. Let $g : \beta(X) \rightarrow \{0, 1\}$ be the extension of f . Then $g^{-1}(\{0\}), g^{-1}(\{1\})$ is a separation of $\beta(X)$.

Let X be a metrizable space. X has a metrizable compactification iff X is second countable. If X is regular second countable, embed $X \xrightarrow{f} [0, 1]^\omega$. The compactification Y induced by f is homeomorphic to a subspace of the metrizable space $[0, 1]^\omega$, so Y is metrizable. Conversely, suppose Y is a metrizable compactification of X . Since Y is compact, Y is second countable. Since X is a subspace of Y , X is also second countable.

Let X have the discrete topology. Then $\beta(X)$ is totally disconnected. Suppose $X = C \cup D$, C, D are arbitrary disjoint. Then $f : X \rightarrow \{0, 1\}$ st $f(C) = \{0\}$ and $f(D) = \{1\}$ is continuous. Let $g : \beta(X) \rightarrow \{0, 1\}$ be the extension of g , so $C \subset g^{-1}(\{0\})$ and $D \subset g^{-1}(\{1\})$. Hence (closures wrt $\beta(X)$), $\overline{C}, \overline{D}$ are disjoint. Since $\beta(X) = \overline{X} = \overline{C \cup D} = \overline{C} \cup \overline{D}$, $\overline{C}, \overline{D}$ form a separation of $\beta(X)$ (if we take $C, D \neq \emptyset$), so any two points within X are separated wrt $\beta(X)$.

Now consider $x \in X$. Since $\{x\}$ is open wrt X , there exists $U(x) \subset \beta(X)$ st $U \cap X = \{x\}$. Sps FTSOC there exists $y \in (\beta(X) - X) \cap U$. Since $\beta(X)$ is Hausdorff, take $N(y)$ disjoint to x . Then $N \cap U$ is a neighborhood of y disjoint

to X , contradiction as $y \in \beta(X) = \overline{X}$. Hence, $U \subset X$, so $U = \{x\}$ is open, so a point in X is separated from a point in $\beta(X) - X$.

Suppose $y_1, y_2 \in \beta(X) - X$ are distinct. Pick disjoint $N_1(y_1), N_1(y_2)$. Let $C_1 = N_1 \cap X \neq \emptyset$ and $C_2 = N_2 \cap X \neq \emptyset$. There exist disjoint D_1, D_2 st $X = D_1 \cup D_2$ and $D_1 \supset C_1, D_2 \supset C_2$. Consider any open $U_1(y_1)$. $U_1 \cap N_1$ is a neighborhood of y_1 and intersects $C_1 \subset D_1$, so $y_1 \in \overline{D_1}$. Similarly, $y_2 \in \overline{D_2}$. $\overline{D_1}, \overline{D_2}$ is a separation of $\beta(X)$, so any two points of $\beta(X) - X$ are separated.

**Show that any continuous real function defined on S_Ω is eventually constant. Counterexample?

Let N have the discrete topology. Then $|\beta(N)| = |I^I|$. There are at most $|J| \leq |\mathbb{R}^N| = |\mathbb{R}|$ bounded functions $N \rightarrow \mathbb{R}$. Since $\beta(N) \subset \mathbb{R}^J$, $|\beta(N)| \leq |\mathbb{R}^J| = |I^I|$. Now I^I has a countable dense subset D . Let $f : N \rightarrow D$ be a continuous bijection (not homeomorphism), so the extension $f : N \rightarrow D$ is also continuous. Since I^I is compact Hausdorff, let $g : \beta(N) \rightarrow I^I$ be the extension of f . Note $g(\beta(N)) \supset g(N) = f(N) = D$. Since $\beta(N)$ is compact, so is $g(\beta(N))$, so $g(\beta(N))$ is closed wrt I^I . Now

$$I^I = \overline{D} \subset \overline{g(\beta(N))} = g(\beta(N)) \subset I^I,$$

so $g(\beta(N)) = I^I$. Since g is surjective, $|\beta(N)| \geq |I^I|$, so $|\beta(N)| = |I^I|$.

Let X be normal and $y \in \beta(X) - X$. Show that y is not the limit of a sequence of points in X . Suppose $(x_n) \subset X$ converges to $x_\infty = y$. (x_n) has only finitely many instances of each x_i , otherwise (x_n) converges to an element of X . Hence, we may pick a subsequence (y_n) of (x_n) consisting of different terms.

The only limit point of (y_n) is $y_\infty = y$, so $\{y_n\} \cup \{y_\infty\} \subset \beta(X)$ is closed, so $\{y_n\} \subset X$ is closed and inherits the discrete topology. Define continuous $f : \{y_n\} \rightarrow I$ by $f(y_n) = n\%2$. Since X is normal and $\{y_n\}$ is closed, extend by Tietze extension theorem f to $g : X \rightarrow I$. Since I is compact Hausdorff, extend g to $h : \beta(X) \rightarrow I$, but h can't be continuous: let $U = (h(y) - 1/2, h(y) + 1/2)$. $f^{-1}(U)$ can't contain any neighborhood of y .

If X is completely regular and noncompact, then $\beta(X)$ is not metrizable. Assume $\beta(X) \neq X$. If $\beta(X)$ is metrizable, then pick $y \in \beta(X) - X$, so there exists a sequence $(x_n) \subset X$ converging to y (since $y \in \overline{X}$). Hence, X is not normal, so X is not compact.